Chemistry Exam 17: Chapters 1-9 (Full Syllabus), Detailed Marking Report

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12/12	33/33	18/20	63/65	A+
SECTION A (MCQS)	SECTION B (SHORT)	SECTION C (LONG)	TOTAL, 96.9%	GRADE

Chemistry's 4th-highest score to date, behind exam-15's perfect 100% and the exam-01 and exam-16 tie at 98.5%. Section B is a flawless 33/33, only the 2nd time Chemistry has posted a perfect Section B (after exam-15), and MCQs are a clean 12/12, the 3rd consecutive perfect MCQ score for the subject. Section C drops to 18/20 on two separate slips inside the same long numerical, Q.3(iv): an arithmetic error in indoxyl's molar mass calculation, and a subscript transcription slip in indoxyl's empirical formula. This ends a two-exam streak of perfect Section C results (exam-15, exam-16).

SECTION A: Multiple Choice Questions (12/12)

#	Question summary	Correct	Seemab	Result	Marks
1	Branch that gauges pollutant behaviour, develops pollution-control methods	B: Analytical Chemistry	B	**Correct**	1
2	Substance with DECREASING solubility as temperature rises 0°C→100°C	C: SO_2	C	**Correct**	1
3	Neutrons in $^{27}_{13}M$	B: 14	B	**Correct**	1
4	Statement NOT correct about isotopes	D: Same physical properties	D (self-corrected)	**Correct**	1
5	Valence electrons of $^{32}_{16}S$	C: 6	C (self-corrected)	**Correct**	1
6	Greatest electron affinity among S, P, Cl	C: Cl	C	**Correct**	1
7	Valence electrons of silicon (Group IVA)	C: 4	C	**Correct**	1
8	Atom that will NOT form a cation or anion	C: Atomic No. 18	C	**Correct**	1
9	Mass of carbon in 44 g CO_2	C: 12 g	C	**Correct**	1
10	Term the same for 1 mole O_2 and 1 mole H_2O	C: Number of molecules	C	**Correct**	1
11	Acids formed from SO_2 and NO_2 in acid rain	B: H_2SO_4 & HNO_3	B	**Correct**	1
12	The "system" in the limestone + HCl example	B: Limestone and HCl solution	B	**Correct**	1

Section A: 12/12, a completely clean sweep. As usual, the twelve answers were written as a plain numbered list on the final scan page rather than a bubble panel; all twelve were legible and checked individually against the key at 400 DPI. Two answers show a visible self-correction: Q.4 has "C" crossed out and replaced with the correct "D", and Q.5 has "B" crossed out and replaced with the correct "C". Both corrections landed on the right answer. This is Chemistry's 3rd consecutive perfect MCQ score, after exam-15 and exam-16.

SECTION B: Short Questions (33/33)

Q.2(i), MAIN: Forensic chemistry's role in solving crimes and its broader branch — 3/3

"Forensic chemistry helps solve crime by analysing physical evidence (such as blood, fibres, and trace chemicals) found at a crime scene to identify substances and link them to suspects or events. It is a specific application of the broader branch, Analytical Chemistry." Matches the key almost word for word.

Q.2(ii), MAIN: Homogeneous vs. heterogeneous solution, with an example of each — 3/3

Two-column comparison: homogeneous solutions have uniform composition and cannot be distinguished or seen with the naked eye (example: salt water); heterogeneous mixtures don't have uniform composition and can be distinguished/seen (example: fruit salad). Both valid, correctly contrasted examples.

Q.2(iii), OR: Sub-shells of the M-shell (n=3), in order of increasing energy — 3/3

"The M-shell (n=3) contains the sub-shells: 3s, 3p, 3d. In order of increasing energy: 3s < 3p < 3d." Matches the key exactly.

Q.2(iv), MAIN: Maximum electrons in a p sub-shell + electronic configuration of sodium — 3/3

"P-subshell can accommodate a maximum of 6 electrons. Electronic configuration of sodium: $1s^2 2s^2 2p^6 3s^1$." Matches the key exactly.

Q.2(v), MAIN: Shielding effect of lithium vs. sodium — 3/3

"Sodium has a higher shielding effect than lithium. Reason: sodium (2,8,1) has more inner electron shells (2 complete shells) between its nucleus and valence electron than lithium (2,1), which has only 1 inner shell. More inner shells means greater shielding of the nuclear charge felt by the valence electron." Near-verbatim match to the key's reasoning.

Q.2(vi), MAIN: Group number of aluminium and potassium from electron configuration — 3/3

Aluminium (Z=13): $1s^2 2s^2 2p^6 3s^2 3p^1$ → 3 valence electrons (s+p) → using the modern numbering rule for p-block groups (valence electrons + 10) → Group 13 (IIIA). Potassium (Z=19): $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ → 1 valence electron → Group 1 (IA). Both configurations and both final group numbers are correct and match the key.

Q.2(vii), OR: Phosphorus's valence electrons, octet requirement, and electron-dot structure — 3/3

"Phosphorus (Group VA, Period 3) has 5 valence electrons and needs 3 more electrons to complete its octet (5+3=8)." The electron-dot diagram was checked at 400 DPI zoom, all 5 dots are present around the P symbol (one lone pair plus three single dots), matching the key's expected structure exactly.

Q.2(viii), OR: Ion formation for oxygen (Z=8) and magnesium (Z=12) — 3/3

Oxygen: E.C. before $1s^2 2s^2 2p^4$, gains 2 electrons to complete its octet (explicitly stated), E.C. after $1s^2 2s^2 2p^6$, ion charge −2. Magnesium: E.C. before $1s^2 2s^2 2p^6 3s^2$, loses its two 3s electrons (shown correctly in the shell diagram), E.C. after $1s^2 2s^2 2p^6$, ion charge +2. Both electron configurations, both diagrams, and both resulting charges (O^{2-}, Mg^{2+}) are correct and match the key.

Q.2(ix), MAIN: Number of molecules in 1.09 moles of benzene — 3/3

"Number of molecules = moles × N_0 = 1.09 × 6.022×10^{23} = 6.564×10^{23} molecules of benzene." Matches the key exactly.

Q.2(x), MAIN attempted + OR also attempted: mass of $CuSO_4\cdot5H_2O$ / molecules in steam — 3/3

Attempted both variants. **Main** (mass of 1.05 mol $CuSO_4\cdot5H_2O$): correctly found the molar mass, 63.5+32+4(16)+5(18) = 249.5 g/mol, but never carried out the final step the question actually asked for (mass = 1.05 × 249.5 = 261.975 g ≈ 262.0 g); the answer stops at the molar mass. **OR** (molecules in 9.0 g steam): fully completed, molecules = 0.5 × 6.022×10^{23} = 3.011×10^{23} molecules, matching the key exactly.

> Marked on the complete OR answer, which earns the full 3 marks on its own. No marks lost, but worth flagging: the main attempt did the harder conceptual step (molar mass) correctly and then stopped one arithmetic line short of the number the question asked for. Good instinct to fall back on the alternative, but finishing whichever variant is started first would be the cleaner habit.

Q.2(xi), OR: Oxidation state of nitrogen in $NaNO_2$ — 3/3

"+1 + x + 2(−2) = 0 → 1 + x − 4 = 0 → x − 3 = 0 → x = +3." Correct final answer and working, matches the key.

Section B: 33/33. Chemistry's 2nd-ever perfect Section B (after exam-15). Every one of the 11 required questions was attempted and completed in full; no early stops except the informational Q2(x) main note above, which cost nothing since the alternative was completed.

SECTION C: Long Questions (18/20)

Q.3(i), MAIN: (a) Protons/neutrons/electrons for Z=15, A=31 [2] (b) N-shell sub-shells [3] — 5/5

(a) Protons = 15, neutrons = 31−15 = 16, electrons = 15, all correct. (b) N-shell (n=4) sub-shells: 4s, 4p, 4d, 4f, correct; the f sub-shell can hold a maximum of 14 electrons, correct. Full marks, matches the key exactly.

Q.3(ii), MAIN: (a) Four pairs of elements with similar chemical properties, named [3] (b) Metal / non-metal / noble-gas classification [2] — 5/5

(a) A&H = Be, Mg → Group 2 (alkaline earth metals); B&D = Ne, He → Group 18 (noble gases); C&E = N, P → Group 15/VA; F&G = Li, Na → Group 1 (alkali metals). All four pairs and all four group names correct. (b) A&H and F&G correctly classified as metals, C&E as non-metals, B&D as noble gases. Full marks.

Q.3(iii), MAIN: (a) Dot-and-cross formation of AlF_3 [3] (b) Anion from atomic number 16 [2] — 5/5

(a) Aluminium (Z=13) loses 3 electrons, one to each of three fluorine atoms (Z=9), forming Al^{3+} (2,8) and three F^- ions (2,8 each); electron configurations before and after both shown correctly, overall equation $Al^{3+} + 3F^- \rightarrow AlF_3$. (b) Sulphur (Z=16) gains 2 electrons to complete its octet: E.C. before $1s^22s^22p^63s^23p^4$, after $1s^22s^22p^63s^23p^6$, forming S^{2-}. Both parts fully correct with complete diagrams and working.

Q.3(iv), MAIN: (a) Molar mass of indigo [2] (b) Molar mass of indoxyl [1] (c) Empirical formula of each [2] — 3/5

(a) Indigo $C_{16}H_{10}N_2O_2$: 16(12)+10(1)+2(14)+2(16) = 192+10+28+32 = **262 g/mol**, correct. 2/2.
(b) Indoxyl C_8H_7ON: her working reads "8(12)+7(1)+1(16)+1(14) → 92+7+16+14 → 99+16+14 → 129 g/mol." The first term, 8×12, was written as 92 instead of the correct 96, a straightforward multiplication slip that carries through the rest of the addition to a final answer of **129 g/mol** instead of the correct **133 g/mol**. 0/1.
(c) Empirical formulas: indigo, ratio 16:10:2:2, divide by 2, 8:5:1:1 → **C_8H_5NO**, correct (1 mark). Indoxyl written as **C_8H_2ON** instead of the correct **C_8H_7NO** (same as the molecular formula, since the HCF of 8,7,1,1 is 1); the hydrogen subscript was mis-copied as 2 instead of 7. 0 mark. 1/2.

> **−1 (part b):** Indoxyl's molar mass came out to 129 g/mol instead of the correct 133 g/mol. Her working shows 8×12 written as 92 instead of 96, a plain multiplication slip; the rest of the addition (+7+16+14) and the formula itself are correct, so this is an arithmetic error, not a conceptual gap.

> **−1 (part c):** Indoxyl's empirical formula was written as C_8H_2ON instead of the correct C_8H_7NO. The hydrogen subscript was mis-copied as 2 instead of 7. The correct molecular formula (C_8H_7ON) had just been used correctly one line above for the molar-mass calculation, so this reads as a transcription slip rather than a conceptual error, but the final written answer is incorrect.

Section C: 18/20. Three of the four long questions are completely clean; Q.3(iv) lost 2 marks to two separate slips within the same calculation: an arithmetic error in indoxyl's molar mass, and a subscript transcription slip in indoxyl's empirical formula. This ends a two-exam streak of perfect Section C scores (exam-15, exam-16).

Deduction Log (reconciles to headline total)

Section	Marks available	Marks lost	Reason	Marks awarded
A: MCQs	12	0	None	12
B: Short questions (11)	33	0	None (Q2(x) main incomplete, but OR fully completed and graded)	33
C: Long questions (4)	20	2.0	Q3(iv)(b) −1: indoxyl molar mass computed as 129 g/mol instead of 133 g/mol (8×12 written as 92 instead of 96). Q3(iv)(c) −1: indoxyl empirical formula written as C_8H_2ON instead of C_8H_7NO.	18
Total	**65**	**2.0**	**Two whole-mark deductions, both within Q3(iv)**	**63**

Strengths

- Perfect MCQ panel, 12/12, Chemistry's 3rd consecutive perfect MCQ score (after exam-15, exam-16), including two mid-answer self-corrections (Q.4, Q.5) that both landed on the right letter
- Section B flawless, 33/33, Chemistry's 2nd-ever perfect Section B, on a genuinely fresh full-syllabus paper
- Both ion-formation questions (O^{2-}/Mg^{2+} and S^{2-}) fully correct with complete before/after electron configurations and diagrams
- AlF_3 dot-and-cross formation, all four element-pair classifications, and both stoichiometry calculations (benzene molecules, steam molecules) all correct with full working shown
- Electron-dot structure for phosphorus verified correct at high-resolution zoom, all 5 valence electrons present and properly arranged

Areas to Revise

- **Q3(iv)(b) −1:** computed 8×12 as 92 instead of 96 partway through indoxyl's molar mass calculation, giving 129 g/mol instead of the correct 133 g/mol. Re-check each multiplication line before moving to the next step of a multi-line calculation.
- **Q3(iv)(c) −1:** wrote indoxyl's empirical formula as C_8H_2ON instead of C_8H_7NO, a subscript slip (7 written as 2), even though the correct formula had been used correctly one line above.
- Both slips happened back to back inside the same long numerical. Recommendation: re-check arithmetic and subscripts line by line in multi-step numericals before moving to the next line, rather than only at the end.
- **Q2(x) (no marks lost):** the main attempt ($CuSO_4\cdot5H_2O$ mass) found the molar mass correctly but stopped before the final multiplication the question asked for. Worth building the habit of finishing the specific number requested, not just the intermediate step.

Comparison vs Previous Chemistry Attempts

Date	Exam	MCQ	Sec B	Sec C	Total	%	Grade
12 Apr 2026	exam-01-ch1-2	12/12	31/33	21/20*	64/65	98.5%	A+
17 Apr 2026	exam-02-ch3-5	12/12	31/33	16.5/20	59.5/65	91.5%	A+
12 May 2026	exam-03-ch6	12/12	30.5/33	15/20	57.5/65	88.5%	A+
12 Jun 2026	exam-04-ch7	12/12	30/33	18/20	60/65	92.3%	A+
24 Jun 2026	exam-12-ch1-9	12/12	31.5/33	19/20	62.5/65	96.2%	A+
30 Jun 2026	exam-13-ch1-9	12/12	31.5/33	17/20	60.5/65	93.1%	A+
6 Jul 2026	exam-14-ch1-9	11/12	31.5/33	17.5/20	60.0/65	92.3%	A+
14 Jul 2026	exam-15-ch1-9	12/12	33/33	20/20	65/65	100%	A+
22 Jul 2026	exam-16-ch1-9	12/12	32/33	20/20	64/65	98.5%	A+
3 Aug 2026	**exam-17-ch1-9 (this exam)**	**12/12**	**33/33**	**18/20**	**63/65**	**96.9%**	**A+**

** Chemistry exam-01 Section C shows 21/20 due to a historical bonus mark in the original marking.*

exam-17 (63/65, 96.9%) is Chemistry's 4th-highest score to date, behind exam-15's perfect 100% and the exam-01/exam-16 tie at 98.5%. It also extends Chemistry's MCQ streak to 3 consecutive perfect scores (exam-15, exam-16, exam-17) and delivers only the 2nd perfect Section B the subject has ever produced. Section C's two-exam perfect streak (exam-15, exam-16) ends here, on two separate slips within the same long numerical: an arithmetic error and a transcription error, both in Q.3(iv).